

SUMMARY (narrative by template; all figures from the deterministic engine)

The request for Rs 90,000 over 36 months is declined under the default policy (v1.0). Verified monthly income is Rs 39,150 against existing obligations of Rs 2,500; the proposed EMI of Rs 3,076 gives a FOIR of 14.2% (limit 50.0%) and an LTV of 75.0% (limit 80.0%). Risk band: LOW. Reason codes: LOW_AGE.

For the borrower:

Ravi Kumar, we could not approve the loan of Rs 90,000 as requested because you are below the minimum age for this product.

Key drivers:

- FOIR 14.2% vs limit 50.0%
- LTV 75.0% vs limit 80.0%
- Verified income Rs 39,150 (volatility 4.1%)
- LOW_AGE: Borrower age 19 is below policy minimum 21

Next steps:

- Re-apply once the borrower meets the minimum age, or add an eligible co-applicant.
- Share the plain-language explanation with the borrower.

1. DECISION

Decision: DECLINED Confidence: 0.82

Reason codes: LOW_AGE

- [blocker] LOW_AGE: Borrower age 19 is below policy minimum 21

1a. PATH TO APPROVAL (automatic what-if on the immutable base run)

No change to tenure (up to the policy maximum) or loan amount (down to -50%) makes this file approvable; the blocker is not purely affordability.

1b. WHY THIS DECISION (policy margins + SHAP; words by template)

Ravi Kumar's loan is declined because a hard policy rule failed: age 19 vs minimum 21 (-2 years).

No change to tenure or loan amount alone makes the file approvable, so the borrower needs higher verified income or lower existing obligations.

The model's risk band is LOW (score 0.2611 on a 0-1 scale where 0 is safest; a typical synthetic borrower scores 0.437336).

What pushes risk up: ltv (loan / asset value) (75.0% vs a typical 68.1%, +0.012); repayment history score (70.0% vs a typical 71.7%, +0.003).

What pulls risk down: foir (obligations / income) (14.2% vs a typical 41.1%, -0.081); emi-to-income ratio (7.9% vs a typical 34.4%, -0.040); employment stability (95.9% vs a typical 71.8%, -0.024).

The risk score is a secondary signal; the policy rules above decide the outcome.

Mathematics:

PASS FOIR 14.2% vs limit 50.0% (-35.8 pp under)

PASS LTV 75.0% vs limit 80.0% (-5.0 pp under)

FAIL age 19 vs minimum 21 (-2 years)

PASS verified monthly income Rs 39,150

risk score 0.261123 = base 0.437336 + sum SHAP -0.176213 (method additive_exact, additivity error 0.0)

phi(foir) = -0.0806 [value 14.2% vs baseline 41.1%]

phi(emi_to_income) = -0.0398 [value 7.9% vs baseline 34.4%]

phi(employment_stability) = -0.0241 [value 95.9% vs baseline 71.8%]

phi(income_volatility) = -0.0183 [value 4.1% vs baseline 22.4%]

phi(income_mismatch_percentage) = -0.0124 [value 0.3% vs baseline 12.7%]

2. BORROWER SUMMARY (declared facts)

ref_id: REF-APP-AGE-009

full_name: Ravi Kumar

age: 19

location_tier: tier_2

employment_type: gig_platform

declared_monthly_income: 39,000.00

loan_request.amount: 90,000.00
loan_request.down_payment: 30,000.00
loan_request.tenure_months: 36
loan_request.vehicle_price: 120,000.00

3. EXTRACTED FACTS (evidence with citations)

full_name = Ravi Kumar conf 0.95 [application_form p.None] auto
monthly_income = 39,000.00 conf 0.95 [application_form p.None] auto
loan_amount = 90,000.00 conf 0.95 [application_form p.None] auto
vehicle_price = 120,000.00 conf 0.95 [application_form p.None] auto
full_name = Ravi Kumar conf 0.85 [kyc.txt p.1] auto
identity = XXXX-XXXX-4321 conf 0.9 [kyc.txt p.1] auto
address = 14 MG Road, Pune 411001 conf 0.85 [kyc.txt p.1] auto
monthly_income = 38,900.00 conf 0.92 [bank_statement.txt p.1] auto
bank_credits = [38500.0, 41000.0, 36200.0, 39800.0, 37900.0, 40100.0] conf 0.9 [bank_statement.txt p.1] auto
existing_monthly_obligations = 2,500.00 conf 0.92 [bank_statement.txt p.1] auto
address = 14 MG Road, Pune 411001 conf 0.85 [bank_statement.txt p.1] auto
vehicle_price = 120,000.00 conf 0.92 [dealer_invoice.txt p.1] auto
loan_amount = 90,000.00 conf 0.92 [dealer_invoice.txt p.1] auto
platform_income = 39,000.00 conf 0.92 [platform_earnings.txt p.1] auto
monthly_income = 39,000.00 conf 0.92 [platform_earnings.txt p.1] auto

4. CALCULATED VALUES (deterministic engine, formula v1)

verified_income: 39,150.00
existing_obligations: 2,500.00
emi: 3,075.99
foir: 0.1424
ltv: 0.75
income_volatility: 0.0408
income_stability: 0.9592
loan_to_income: 0.1916
emi_to_income: 0.0786
input.annual_rate: 14.00
input.existing_monthly_obligations: 2,500.00
input.loan_amount: 90,000.00
input.monthly_incomes: [38500.0, 41000.0, 36200.0, 39800.0, 37900.0, 40100.0]
input.tenure_months: 36
input.vehicle_price: 120,000.00

5. RISK SIGNAL (secondary, never decides)

score 0.2611 band LOW model xgb-v1/fallback

6. AI REASONING

strengths: FOIR 14% is comfortably serviceable
strengths: LTV 75% leaves borrower equity cushion
risks: INCOME_MISMATCH: Monthly income differs across documents by 0.3%
reasoning: Deterministic FOIR 14.24%, LTV 75.00%; XGBoost band LOW (score 0.2611).

7. POLICY

profile default version 1.0
limit.foir_max: 0.5
limit.ltv_max: 0.8
limit.max_tenure_months: 60
limit.min_age: 21

8. HUMAN OVERRIDES

none

SYNTHETIC DATA - demo only. Financial numbers computed deterministically; LLM never sets numbers.